Identification of Unknown “B”

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Introduction

The purpose is to identify unknown compound “B” using data from Mass Spectroscopy (MS), Nuclear Magnetic Resonance (NMR), and Infrared Spectroscopy (IR). There are multiple methods used in organic chemistry to analyze unknown compounds so that what they are can be determined. Of the various methods and techniques available, MS, NMR, and IR are the most common and reliable forms that allow for individuals in organic chemistry to distinguish between compounds with the same chemical formula but different structural formula.

Body

Mass Spectrometry (MS)

Mass Spectrometry is a technique used in organic chemistry to identify a compound based on mass and energy. To decipher the chemical composition of the compound, the mass to charge ratio is analyzed. There are five key steps involved. The first step involves turning the initial sample (can be a liquid, solid, or gas) into ions. This is known as ionization. Once the sample is ionized, the ions are accelerated by an electric field and focused into a beam. This step is known as acceleration, and it gives all of the ions in the sample the same kinetic speed. Once there is a universal speed observed with the ions, they are passed through a magnetic field. Passing them through the magnetic field bends the ions. The lighter ions also described as the ions with a higher charge are deflected more than the heavier ions or ions with a lower charge. It is important to note that there are several different types of mass analyzers. Different types of detectors measure how much time it takes for particles to reach the detector, how much is deflected, and can yield information besides just mass. After deflection is completed, the ions are separated. The sorting of them is done based on their mass to charge ratio. This is referred to as detection. Detection is achieved by recording the current brought upon by the ion striking a

surface. Finally, once detection is complete, the data accumulated from the process can be analyzed. When analyzing the spectrum, the peaks represent the various ions present in the sample.

Mass Spectrometry can be used to analyze chemical compounds both qualitatively and quantitatively. There are many different things it can identify with the most common ones being the identification of elements in a sample, mass, and chemical structures. Like any technique, there are pros and cons associated with MS. In terms of advantages, the biggest one is that MS is highly sensitive making it a great tool to identify an unknown component of a sample. However, in regards to the disadvantages, MS does not do a great job at identifying hydrocarbons. It also cannot decipher between optical and geometrical isomers.

Nuclear Magnetic Resonance (NMR)

Nuclear Magnetic Resonance is commonly used in organic chemistry to find the structure of a compound, determine molecular dynamics, and study both the relationship between molecules based on their interactions. In order for NMR to work properly, it must rely on what is known asd nuclear spin. Nuclear spin can be defined as the total angular momentum of a nucleus. When nuclei spin, they generate a magnetic moment which can then be picked up by a magnetic field. In terms of how NMR works, a sample is placed in a strong magnetic field creating an alignment of the nuclei either parallel or antiparallel to the field. The use of a radiofrequency pulse is directed perpendicular to the magnetic field which causes the nuclei to flip into the opposite alignment position; however, when the pulse is turned off, the nuclei return to their normal alignment state. When the nuclei are relaxing, they put off radiofrequency signals also known as Larmor frequency. These signals are the differences in energy levels of the nuclear

spin states. These signals are detected and recorded by the radiofrequency coils located inside of the NMR instrument. The final spectrum is generated by Fourier which transforms the signals.

There are various NMR parameters to highlight. Chemical shift indicates the chemical environment the nucleus is in. Coupling constants provide data about neighboring nuclei. Signal intensity describes how many nuclei contribute to each individual signal. NMR is a great technique for identifying structural information of a compound. It is also non-destructive, versatile, and provides high resolution results. However, NMR can be less sensitive when compared to other analytical techniques which can be a downfall when working with samples of low concentrations or of low quantity.

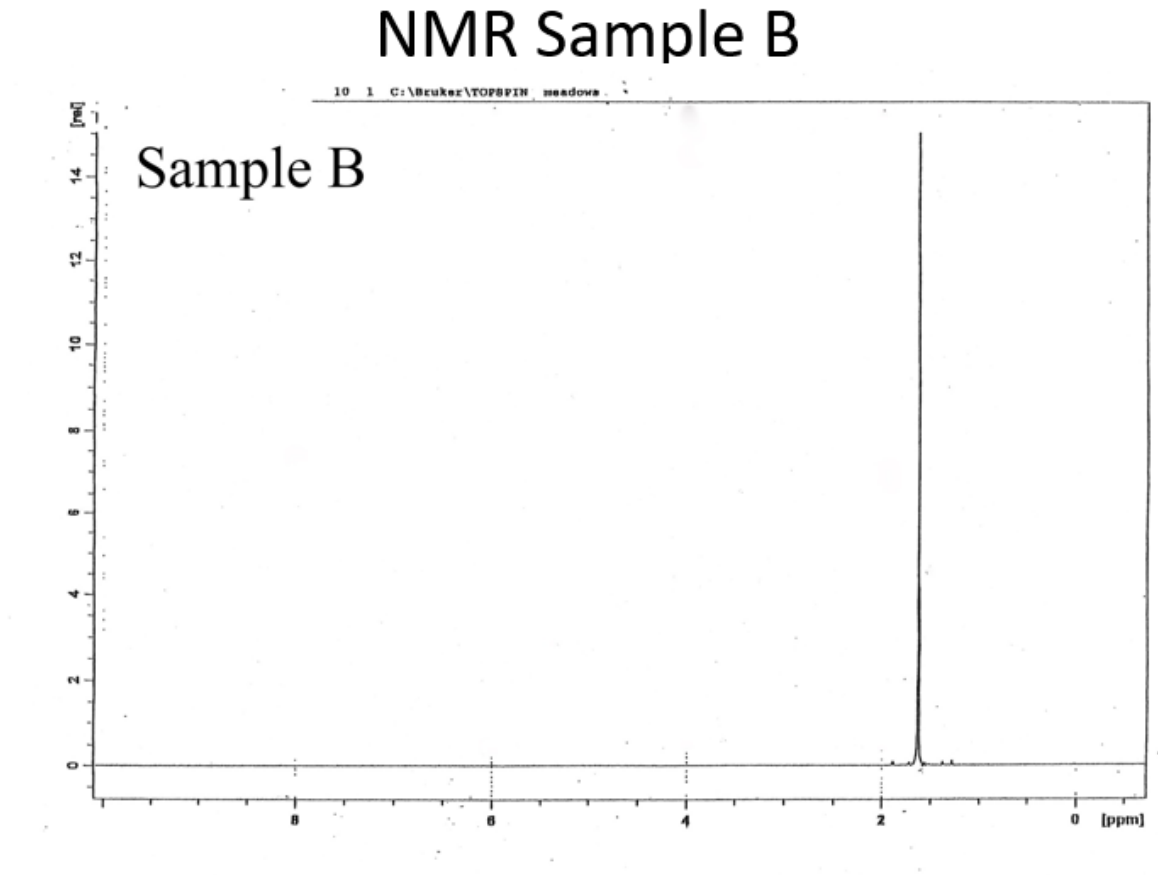
Infrared Spectroscopy (IR)

In organic chemistry, infrared spectroscopy is a technique that measures the absorption of infrared radiation by a chosen sample. The absorption is done via the vibrational modes between the chemical bonds within a molecule. Infrared radiation deals with electromagnetic waves. These waves have wavelengths that are longer than visible light. In terms of how IR works, infrared radiation first passes through a sample where some of it will be absorbed by the molecules inside of the sample and some of it will not. The absorption of the infrared radiation results in an absorption spectrum where the absorbed radiation is plotted as wavelength of frequency. Chemical bonds are unique which is why each of them have individual absorption levels leading to specific peaks in the spectrum. IR is a great tool for identifying the specific functional groups present in the sample since the positions and shapes of the peaks correlate to individual functional groups. Not only can IR be used to identify what functional groups are present in the sample, but it can also give information regarding the concentration of the functional groups. This is done by the intensity of the absorption bands.

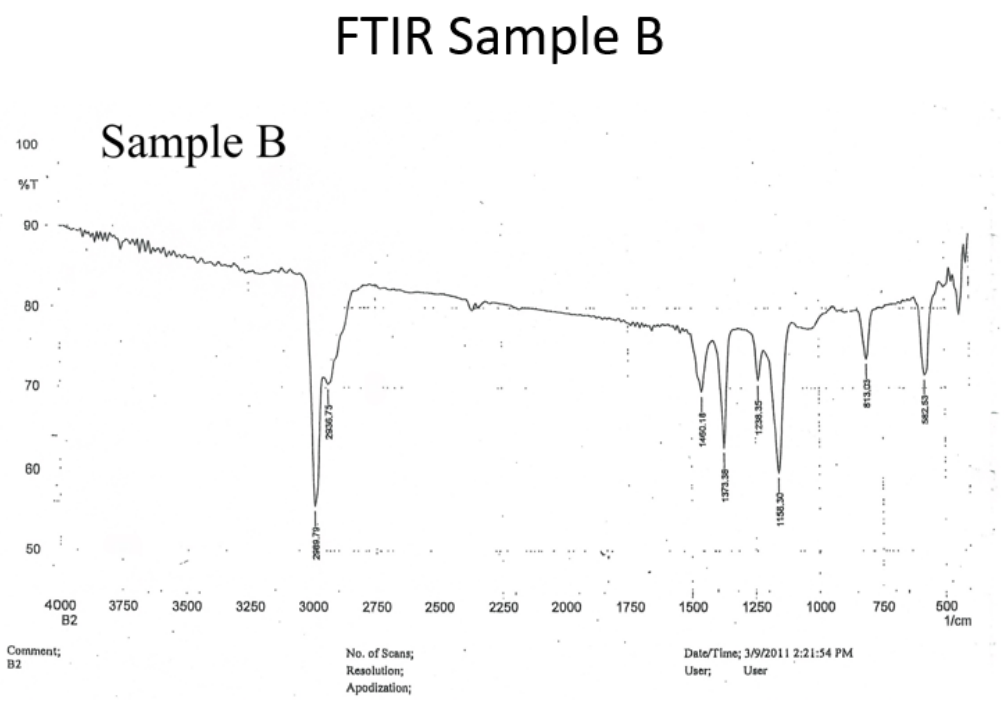
IR is a great analytical technique available to identify functional groups within a sample. It is also highly versatile and non-destructive. IR can be sensitive when working with samples of low concentration or of low quantity. It can also be difficult to read sometimes due to overlapping peaks on the spectrum.

Data

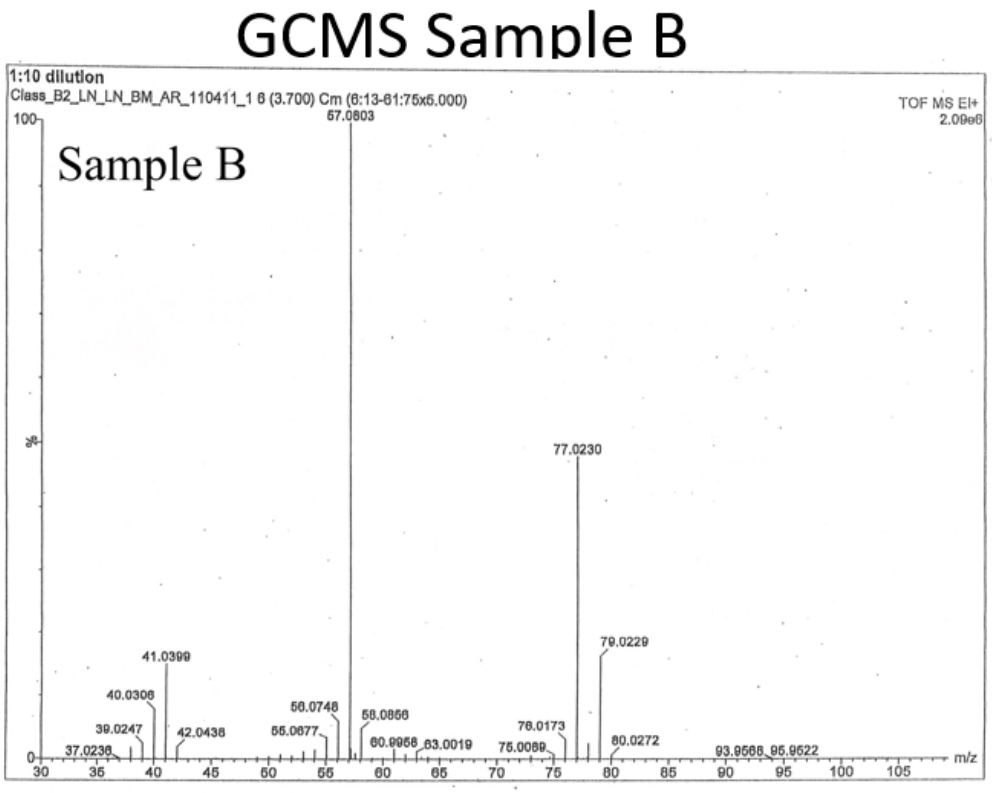
NMR Unknown Sample “B”



IR Unknown Sample “B”



MS Unknown Sample “B”



Analysis

NMR

A single peak at approximately 1.6 ppm is observed. This corresponds with a chloride atom being present in the compound causing a chemical downfield shift. There is only one type of proton present in the compound which is based on there only being one single peak on the graph. Also, since the singular peak is very tall, it can be concluded that there are many protons contributing to the signal.

IR

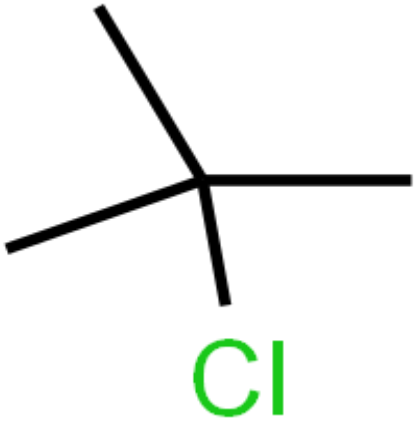
There are no double or triple bonds present in the compound. This is derived from there being no peaks between 1600-1700 cm^{-1}. There also is not an alcohol group present in the compound since there is no noticeable activity above 3000 cm^{-1}. There are C-H bonds present in the compound since there are peaks present at 2989.79 cm^{-1} and 2936.75 cm^{-1}. Methyl groups are present in the compound since there is activity between 1500-100 cm^{-1}. Finally, it can be concluded that there is a C-Cl bond since there are peaks located at 813.03 cm^{-1} and 582.83 cm^{-1}.

MS

The base peak is located at 57 m/z. The $(M+1)^+$ peak is located at 77 m/z and the $(M+2)^+$ peak is located at 79 m/z. The $(M+1)^+$ peak is approximately ⅔ taller than the $(M+2)^+$ peak which corresponds to a chlorine atom being present in the compound. Finally, the molecular ion peak is located at 93 m/z. This indicates that the compound's chemical formula is tert-butylchloride (t-butylchloride).

Conclusion

Three analytical methods were used to determine the composition of the unknown compound. After analyzing the data on the IR, NMR, and MS spectrums, the unknown chemical compound “B” is most probably ***tert-butylchloride***. An image of its structure is shown below.



References

Keller L, Joseph K, Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry 2 Lab Manual [lab manual]. Auburn (Al): Auburn University.

Libretexts. (2023, August 29) How the mass spectrometer works. *Chemistry LibreTexts*.

Libretexts. (2022, April 16) Infrared Spectroscopy. *Chemistry LibreTexts*. Libretexts.

Libretexts. (2023, January 30) Introduction to NMR. *Chemistry LibreTexts*. Libretexts.